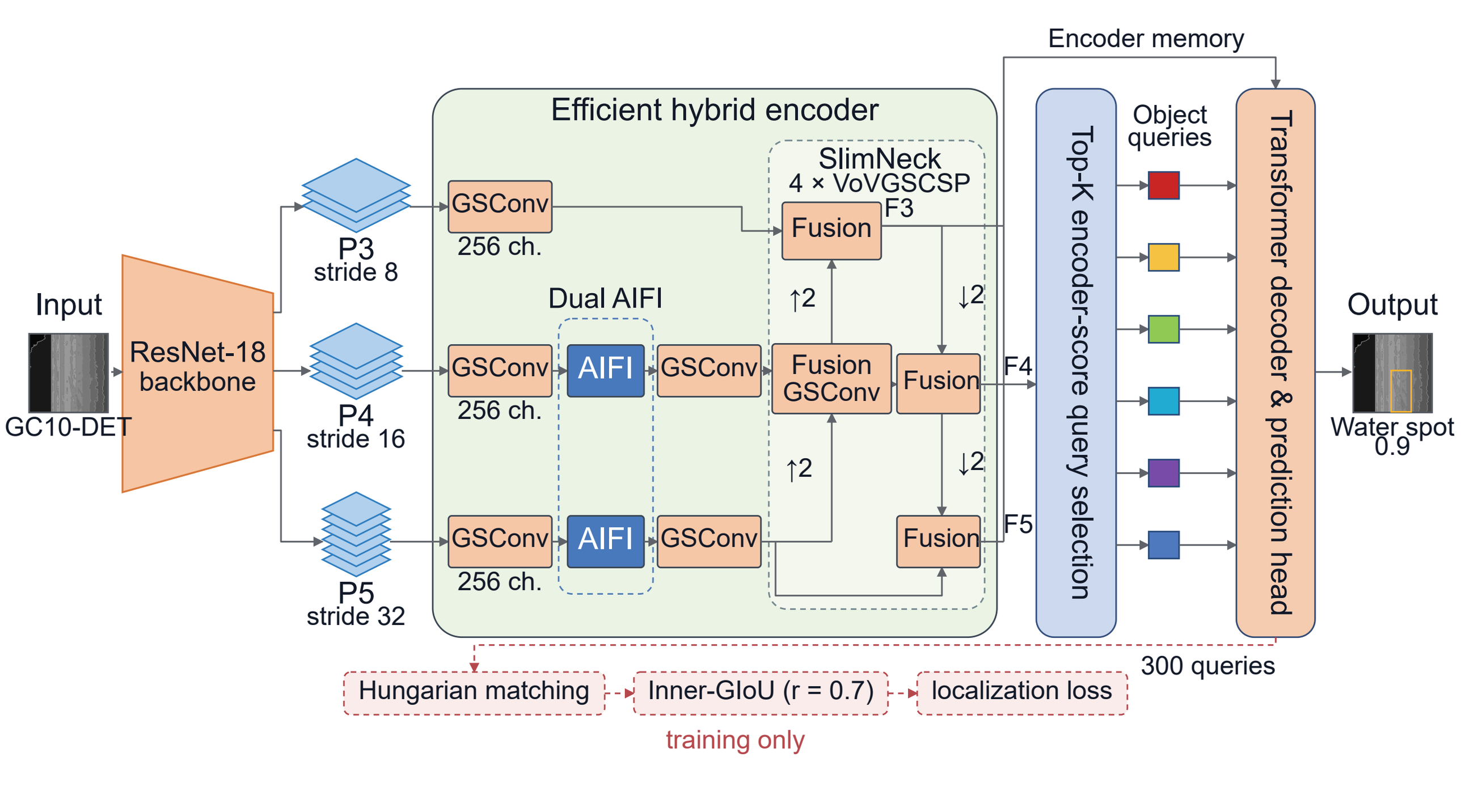

Input
GC10-DET
ResNet-18
backbone
P3
stride 8
P4
stride 16
P5
stride 32
Efficient hybrid encoder
GSConv
256 ch.
GSConv
256 ch.
GSConv
256 ch.
Dual AIFI
AIFI
AIFI
GSConv
GSConv
GSConv
SlimNeck
4 × VoVGSCSP
Fusion
F3
↑2
↓2
Fusion
GSConv
Fusion
F4
↑2
↓2
Fusion
F5
Encoder memory
Top-K encoder-score query selection
Object
queries
Transformer decoder & prediction head
Output
Water spot
0.9
300 queries
Hungarian matching
Inner-GIoU (r = 0.7)
localization loss
training only